

A peak majorant need only commute with the peak projection

Candidate full affirmative answer to a question in arXiv:1206.4022

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Status: candidate full proof, likely valid. Corollary 5.4 of Blecher–Read remains true when its hypothesis that the positive invertible majorant d commute with the peak element a is replaced by the weaker hypothesis $dq = qd$.

1 Source question and exact scope

The source is D. P. Blecher and C. J. Read, *Operator algebras with contractive approximate identities II*, arXiv:1206.4022v4, J. Funct. Anal. 264 (2013), 1049–1067 [1]. Corollary 5.4 assumes that A is an operator algebra in a unital C^* -algebra B , that q is the peak of $a \in \text{Ball}(A^1)$, that $b \in A$ and $|b|$ commutes with a , and that

$$qb^*bq \leq qdq$$

for an invertible $d \in B_+$. Under the additional condition $ad = da$, it produces $x \in A$ with $xq = bq$ and $x^*x \leq d$.

Thus

$$\|gs\| \leq \sum_{k=1}^{n-1} \frac{\|g_k s\|}{2^k} + \sum_{k=n}^{\infty} \frac{1}{2^{k+1}} \leq (1 + \frac{1}{2^n})(1 - \frac{1}{2^{n-1}}) + \frac{1}{2^n} = 1 - \frac{1}{2^{2n-1}} \leq 1.$$

Since $\|g\| = \||b'| \sum_{k=1}^{\infty} \frac{a^{m_k}}{2^k}\|$, and the (u_k) commute with $|b'|$ and a^{m_k} , we see that $\|g\|$ is dominated by the maximum of $\|g(1-u_2)\|$, $\|g(\wedge_k u_k)\|$, and $\sup_{n \geq 3} \|g(u_{n-1} - u_n)\|$, each of which is ≤ 1 . So $g^*g \leq 1$ and $x^*x = fg^*gf \leq f^2 = d$. $\square$

Remarks. 1) Corollary 5.4 Theorem 5.1 and the matching results in Section 2, lead to ‘Rudin-Carleson theorems’ of the type in [21] Proposition 3.4].

2) Corollary 5.4 is not true if one drops the condition that d commutes with a . For example, take A to be the lower triangular matrices in M_2 , $q = a = E_{11}$, and $d = \epsilon E_{22} + |x\rangle\langle x|$, where $x = [1 \ -1]$. We do not know if Corollary 5.4 is true if one replaces the condition that d commutes with a with d commuting with q .

6. COMPACT PROJECTIONS

In a recent paper [12] the first author and Neal developed (generalizing work of Akemann and coauthors, see e.g. [3]) the theory of compact projections in an approximately unital operator algebra A . We defined a projection $a \in A^{**}$ to be

Figure 1: The end of Corollary 5.4 and Remark 2 on source PDF page 14.

The exact question is:

“We do not know if Corollary 5.4 is true if one replaces the condition that d commutes with a with d commuting with q .”

We retain every other hypothesis of the corollary. Products with q below are computed in B^{**} via the canonical normal extension of the inclusion.

2 Proof intuition

Writing $f = d^{-1/2}$ turns $x^*x \leq d$ into $\|xf\| \leq 1$. Normalizing the whole algebra is unhelpful because f need not commute with a . The key is to normalize only the cyclic space generated by b and a :

$$X = \overline{b \text{ oa}(1, a) f}.$$

Let J be the peak ideal in $\text{oa}(1, a)$, consisting of the functions that vanish at q , and put $Y = \overline{bJf}$. The commutation assumptions imply that every q -column from X has range orthogonal to every $(1-q)$ -column from X . Consequently right multiplication by $1-q$ is an M -projection on X^{**} , with range $Y^{\perp\perp}$. Thus Y is an M -ideal and exact best approximation is available. The source inequality says precisely that the quotient norm of bf is at most one.

The fixed orthogonal range spaces in the proof are important: they ensure that the max-norm decomposition survives passage to the bidual, avoiding an invalid appeal to weak-star continuity of products.

3 The cyclic weighted-space lemma

We use the standard characterization that Y is an M -ideal in a Banach space X when a projection $P : X^{**} \rightarrow Y^{\perp\perp}$ satisfies

$$\|z\| = \max\{\|Pz\|, \|z - Pz\|\} \quad (z \in X^{**}).$$

Every M -ideal is proximal [5].

Lemma 1 (Cyclic weighted peak ideal). *Let E be a unital commutative operator algebra contained in a unital C^* -algebra B . Let $q \in E^{**} \subset B^{**}$ be a central closed projection and*

$$J = \{h \in E : hq = 0\}, \quad J^{\perp\perp} = E^{**}(1 - q).$$

*Suppose $b \in B$, $f \in B_+$ is invertible, $fq = qf$, and b^*b commutes with E and q . Then, for*

$$X = \overline{bEf}, \quad Y = \overline{bJf},$$

*Y is an M -ideal in X . Under the canonical realization of X^{**} as the weak-star closure of X in B^{**} , its M -projection is $Pz = z(1 - q)$. In particular,*

$$\text{dist}(z, Y) = \|zq\| \quad (z \in X), \tag{1}$$

and the distance is attained.

Proof. Represent B^{**} normally and faithfully on a Hilbert space H . If $h, k \in E$, then

$$(bhfq)^* bkf(1 - q) = qfh^* b^* bkf(1 - q) = 0, \tag{2}$$

because q commutes with every factor between it and $1 - q$. By norm density, the same holds with bhf, bkf replaced by arbitrary elements of X .

Let H_q be the closed linear span of all ranges xqH , $x \in X$, and let H_0 be the closed linear span of all ranges $x(1 - q)H$, $x \in X$. Equation (2) gives $H_q \perp H_0$. If a bounded net in X converges weak-star to $z \in X^{\perp\perp}$, then its q -columns converge weakly pointwise to zq and its $(1 - q)$ -columns to $z(1 - q)$. Closed subspaces of H are weakly closed; hence

$$zqH \subset H_q, \quad z(1 - q)H \subset H_0.$$

Thus $(zq)^*z(1-q) = 0$, and therefore

$$\|z\| = \max\{\|zq\|, \|z(1-q)\|\} \quad (z \in X^{\perp\perp}). \quad (3)$$

It remains to identify the range. The hypothesis on $J^{\perp\perp}$ gives a bounded net $(e_t) \subset J$ converging weak-star to $1-q$. For $h \in E$, using $fq = qf$,

$$bhf(1-q) = bh(1-q)f = \text{w}^*\text{-}\lim_t bhe_t f \in Y^{\perp\perp}.$$

Norm density and then weak-star density show $X^{\perp\perp}(1-q) \subset Y^{\perp\perp}$. Conversely $Yq = 0$, so right multiplication by $1-q$ fixes $Y^{\perp\perp}$. Hence

$$P(X^{\perp\perp}) = X^{\perp\perp}(1-q) = Y^{\perp\perp}.$$

Together with (3), this says that P is an M -projection and Y is an M -ideal in X . In the decomposition $X^{**} = Y^{\perp\perp} \oplus_\infty X^{**}q$, the quotient norm of $z + Y$ is $\|zq\|$, proving (1). Proximality of M -ideals gives attainment. $\square$

4 Affirmative answer

Theorem 1. *Let $A \subset B$ be an operator algebra, where B is a unital C^* -algebra, and identify $A^1 = A + \mathbb{C}1_B$. Suppose $q \in (A^1)^{**}$ is the peak of some $a \in \text{Ball}(A^1)$, $b \in A$, and $|b|$ commutes with a . If $d \in B_+$ is invertible,*

$$dq = qd, \quad qb^*bq \leq qdq,$$

then there is $x \in A$ such that

$$xq = bq, \quad x^*x \leq d.$$

Thus the answer to the question after Corollary 5.4 is yes.

Proof. Let $E = \text{oa}(1, a) \subset A^1$. Since q is the peak of a , it belongs to E^{**} , is closed there, and $1-q$ is the support projection of the ideal

$$J = \{h \in E : hq = 0\};$$

equivalently $J^{\perp\perp} = E^{**}(1-q)$ (this is the standard peak-ideal correspondence; see [4, 1]).

Put $c = b^*b$ and $f = d^{-1/2}$. The condition that $|b|$ commutes with a implies that c commutes with E . Taking weak-star limits of a^n also shows that c commutes with q . The condition $dq = qd$ gives $fq = qf$. Lemma 1 therefore applies to

$$X = \overline{bEf}, \quad Y = \overline{bJf}.$$

For $z_0 = bf$, the compression hypothesis gives

$$\|z_0q\|^2 = \|qfb^*bfq\| \leq \|qfdfq\| = \|q\| = 1.$$

Here the inequality is conjugation of $qb^*bq \leq qdq$ by qfq , and all insertions of q are legitimate because q commutes with c and f . By (1) and proximality, choose $y \in Y$ with

$$\|bf - y\| = \text{dist}(bf, Y) = \|bfq\| \leq 1.$$

Right multiplication by $f^{-1} = d^{1/2}$ maps Y into $\overline{bJ} \subset A$, because $bA^1 \subset A$. Define

$$x = b - yf^{-1} \in A.$$

Since $Yq = 0$ and f^{-1} commutes with q , we have $xq = bq$. Moreover

$$xf = bf - y,$$

so $\|xf\| \leq 1$. Consequently $fx^*xf \leq 1$, and conjugating by f^{-1} yields $x^*x \leq f^{-2} = d$. $\square$

5 Verification, limitations, and novelty

The proof was checked against four failure modes. First, the source's 2×2 example does not commute with q . Second, the proof never claims that d or f commutes with a . Third, the bidual max decomposition is proved using fixed orthogonal output subspaces, not joint weak-star continuity of multiplication. Fourth, both inclusions in $X^{**}(1 - q) = Y^{\perp\perp}$ are explicit. The accompanying deterministic check tests the mixed-product zero identity and max-norm decomposition in finite matrix models; it is a sanity check, not a substitute for the proof.

The theorem answers exactly the question on source PDF page 14. It retains invertibility of d , the peak hypothesis, and commutation of $|b|$ with a ; it does not assert that any of those assumptions can be removed.

For novelty, bounded searches on 13 August 2026 covered the run indexes, the exact question phrase, its main commutation variants, arXiv:1211.5010 [2], and arXiv:1407.1356 [3]. No later primary source answer was found, and the 2018 arXiv source version still contains the open question. Novelty confidence is moderate rather than definitive because the argument uses standard M -ideal technology and may be implicit elsewhere.

Human-review recommendation. Check Lemma 1, especially the canonical realization of X^{**} in B^{**} , persistence of the fixed range orthogonality, and the range identity for the M -projection. Then check the final right-normalization calculation. These are the only substantive points.

References

- [1] D. P. Blecher and C. J. Read, *Operator algebras with contractive approximate identities II*, J. Funct. Anal. **264** (2013), 1049–1067; <https://arxiv.org/abs/1206.4022>.
- [2] D. P. Blecher, *Noncommutative peak interpolation revisited*, Bull. Lond. Math. Soc. **45** (2013), 1100–1106; <https://arxiv.org/abs/1211.5010>.
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- [5] P. Harmand, D. Werner, and W. Werner, *M-ideals in Banach Spaces and Banach Algebras*, Lecture Notes in Mathematics 1547, Springer, 1993.